Telecom NetOps AI Assistant — REST API Specification

Version: 3.0.0 | **Base URL:** http://127.0.0.1:8000 | **Specification:** OpenAPI 3.1 (/openapi.json) | **Total Endpoints:** 18

Master Directory of All 18 Endpoints

#	Method	Endpoint Path	Category	Core Functionality
1	GET	`/health`	System	Liveness probe, loaded row count, active LLM model
2	GET	`/metadata`	System	Dropdown choices: states, cities, seasons, weather
3	GET	`/stats`	System	Whole-dataset KPI averages, condition breakdown
4	POST	`/search-incidents`	Search	Hybrid BM25 + dense vector ranking with relaxation
5	GET	`/incidents/{incident_id}`	Search	Complete 27-metric historical record lookup
6	POST	`/search-incidents/export`	Search	Export all matching search results to streaming CSV
7	POST	`/diagnose`	Diagnosis	Full 10-Agent pipeline, ML throughput/risk & RAG dossier
8	POST	`/feedback`	Feedback	Operator ratings (1-5 stars) and audit notes to SQLite
9	GET	`/feedback/summary`	Feedback	Summary rating statistics and distribution
10	GET	`/feedback/recent`	Feedback	List recent operator reviews (configurable limit)
11	GET	`/dataset/options`	Explorer	Dataset Explorer circles, severities, issues
12	GET	`/dataset`	Explorer	Server-side paginated and sorted 100K dataset view
13	GET	`/dataset/export`	Explorer	Export filtered dataset explorer rows to CSV
14	POST	`/evaluate/retrieval`	Evaluation	Automated Hit@k keyword test over 100K records
15	GET	`/evaluate/ragas`	Evaluation	Retrieve cached RAGAS benchmark scores
16	POST	`/evaluate/ragas`	Evaluation	Trigger live RAGAS evaluation execution
17	GET	`/evaluate/deepeval`	Evaluation	Retrieve cached DeepEval assertion test results
18	POST	`/evaluate/deepeval`	Evaluation	Trigger live DeepEval assertion test harness

1. System & Health Endpoints

[GET] GET /health

Liveness probe reporting backend status, loaded incident rows, whether Chroma vector embeddings are enabled, active LLM model (e.g. claude-sonnet-4-5-20250929), and demo mode status.

Payload / Spec:
```
Status 200 OK: {"status": "ok", "incident_count": 100000, "llm_provider": "anthropic (claude-sonnet-4-5-20250929)"}
```

[GET] GET /metadata

Returns lists of unique States, Cities, Seasons, Weather categories, and Network Conditions in the dataset for frontend dropdowns.

Payload / Spec:
```
Status 200 OK: {"states": [...], "cities": [...], "seasons": [...], "network_conditions": ["Healthy", "At Risk",
"Degraded"]}
```

[GET] GET /stats

Computes aggregate KPIs across all 100,000 records including avg latency (79.22 ms), throughput (586.57 Mbps), packet loss (6.46%), and condition counts.

Payload / Spec:
```
Status 200 OK: {"avg_latency_ms": 79.22, "avg_throughput_mbps": 586.57, "high_risk_rows": 28412}
```

2. Telemetry Search Endpoints

[POST] POST /search-incidents

Executes hybrid BM25 + dense sentence vector ranking over 100,000 rows. Supports state, city, season, weather, condition filters with automatic nearest-match relaxation.

Payload / Spec:
```
Body: {"query": "high latency", "top_k": 5, "state": "Texas"}
Response: {"results": [...], "total_results": 842, "match_mode": "exact"}
```

[GET] GET /incidents/{incident_id}

Looks up all 27 telemetry metrics for a single cell incident ID (e.g. NETOPS-002743).

Payload / Spec:
```
Response 200: {"incident_id": "NETOPS-002743", "metadata": {...}}
Response 404: {"detail": "Incident not found"}
```

[POST] POST /search-incidents/export

Streams all records matching the search query and filters as a downloadable CSV file attachment (netops_search_evidence.csv).

Payload / Spec:
```
Response 200: Content-Type: text/csv; filename=netops_search_evidence.csv
```

3. AI Diagnosis Endpoints

[POST] POST /diagnose

Executes end-to-end 10-Agent cognitive diagnosis: symptom classification, composite prioritization, ML RandomForest throughput regression (Mbps), fault risk probability (%), root cause isolation, impact assessment, and Claude 3.5/4.5 RAG synthesis dossier.

Payload / Spec:

```
Body: {"query": "A cell has high latency and packet loss", "top_k": 5}
Response: {"classification": {...}, "prioritization": {...}, "risk_prediction": {...}, "performance_prediction":
{...}, "rag_answer": {...}}
```

4. Human-in-the-Loop Feedback Endpoints

[POST] POST /feedback

Saves human operator relevance ratings (1 to 5 stars) and audit commentary for specific incident recommendations into SQLite/JSON storage.

Payload / Spec:

```
Body: {"query": "...", "incident_id": "NETOPS-002743", "rating": 5, "comment": "Accurate root cause"}
```

[GET] GET /feedback/summary

Returns overall feedback count, average rating (e.g. 4.65/5.0), and star distribution breakdown.

Payload / Spec:

```
Response 200: {"total_feedback": 48, "average_rating": 4.65, "ratings_distribution": {"5": 34, "4": 11}}
```

[GET] GET /feedback/recent?limit=50

Returns the most recent operator feedback records up to the specified limit.

Payload / Spec:

```
Query: ?limit=50
Response 200: [{"query": "...", "rating": 5, "timestamp": "..."}]
```

5. Dataset Explorer Endpoints

[GET] GET /dataset/options

Returns unique filter choices for the Dataset Explorer: Circles, Severities (CRITICAL/HIGH/MED/LOW), Seasons, and Issue Types (Backhaul, Packet Loss, Coverage Gap).

Payload / Spec:
```
Response 200: {"circle": [...], "severity": [...], "issue_type": [...], "total_count": 100000}
```

[GET] GET /dataset

Server-side filtered, sorted, and paginated exploration over 100,000 cells. Supports sorting by Highest Risk, Lowest Risk, Throughput, Latency, etc.

Payload / Spec:
```
Query: ?severity=CRITICAL&sort_by=Highest+Risk&page=1&page_size=500
Response: {"total_count": 100000, "filtered_count": 14250, "rows": [...] }
```

[GET] GET /dataset/export

Exports the filtered Dataset Explorer records directly as a CSV file download (telecom_dataset_filtered.csv).

Payload / Spec:
```
Response 200: Content-Type: text/csv; filename=telecom_dataset_filtered.csv
```

6. Evaluation & Benchmark Endpoints

[POST] POST /evaluate/retrieval

Runs automated Hit@k evaluation against standardized test queries (evaluation/test_cases.json) across the 100K dataset.

Payload / Spec:
```
Response 200: {"hit_rate": 1.0, "cases": [{"query": "...", "hit@k": true, "top_score": 0.0722}]}
```

[GET] GET /evaluate/ragas & POST /evaluate/ragas

GET retrieves cached RAGAS benchmark scores (Faithfulness: 88%, Relevancy: 93%, Context Precision: 90%). POST runs live evaluation if configured.

Payload / Spec:
```
Response 200: {"summary": {"faithfulness": 0.88, "response_relevancy": 0.93, "context_precision": 0.90}}
```

[GET] GET /evaluate/deepeval & POST /evaluate/deepeval

GET retrieves cached DeepEval test suite results (100% Pass Rate, 96% Relevancy, 98% Faithfulness). POST executes live assertion suite.

Payload / Spec:
```
Response 200: {"summary": {"pass_rate": 1.0, "avg_relevancy": 0.96, "avg_faithfulness": 0.98}}
```

7. Security Guardrails & Error Handling

[GUARD] Ingress Validation & Guardrails

Every query passes through security filters: length bounds (1-1000 chars), Regex XSS/<script> sanitization, and 70+ telecom ontology keyword validation.

Payload / Spec:
```
HTTP 400 Bad Request: "Out-of-domain query rejected: Request contains no recognizable telecom network terminology."
HTTP 422 Unprocessable Entity: Raised by Pydantic if unexpected fields or invalid types are supplied.
```